

&" NuclearCraft Plutonium Age

Complete Player Guide

Crafting Recipes · Materials · Machines · Combat · Boss

PaperMC 1.21+ · Plugin Version 1.0 · Phase 1–10

&" Material Substitution Reference

Because Minecraft's vanilla crafting system cannot filter items by custom data, NuclearCraft uses vanilla base materials as stand-ins in crafting grids. When you see these materials in a recipe, the plugin checks your actual custom item.

Crafting Grid Substitutions	Represents Custom Item	CMD
Vanilla Material (in recipe)		
Echo Shard	Refined Plutonium Ingot	1103
Yellow Wool	Industrial Fabric	1315
Magma Cream	Radioactive Core	1101
Prismarine Crystals	Raw Plutonium Fragment	1102
Heart of the Sea	Irradiated Heart	1106
Nether Star (1)	Reactor Heart	1109
Nether Star (2)	Titan Core	1107
Pink Petals	Healing Petal	1105
Wheat Seeds	Mutated Seed	1104

Base Materials Used for Machine Blocks

Blast Furnace	Nuclear Smelter block	1201
Smithing Table	Nuclear Forge block	1401
Diamond Pickaxe	Radiation Drill	1108
Leather Helmet/Chest/Legs/Boots	Hazmat Suit pieces	1306–1309
Netherite Helmet/Chest/Legs/Boots	Plutonium Armor pieces	3001–3004
Honey Bottle	Radiation Antidote	1301
Glass Bottle	Radiation Serum	1302
Paper	Ancient Reactor Blueprint	1110
Amethyst Shard (1)	Titan Fragment	1108
Amethyst Shard (2)	Mutated Crystal	1111

&" CMD = CustomModelData — the internal number that makes each item visually unique in-game.

&" All Custom Items

Raw Materials & Components

Raw Plutonium Fragment [Prismarine Crystals · CMD 1102]
Mined from Plutonium Ore (deepest layer). Base processing input.
&" Low radiation while carried

Refined Plutonium Ingot [Echo Shard · CMD 1103]
Output of Nuclear Smelter. Primary crafting material for all plutonium gear.
&" Moderate radiation source

Radioactive Core [Magma Cream · CMD 1101]
Dropped by Irradiated Zombies. Fuel for Nuclear Forge. High radiation.
&" High radiation — dangerous

Irradiated Heart [Heart of the Sea · CMD 1106]
Dropped by Irradiated Zombies. Required for MK-3/4 upgrades and Titan summoning.
&" Pulsing with corrupted energy

Industrial Fabric [Yellow Wool · CMD 1315]
Used to craft and repair Hazmat Suits.
&" Radiation-resistant weave

Mutated Seed [Wheat Seeds · CMD 1104]
Dropped by zombies. Plant on Radioactive Farmland to grow Healing Plants.
&" Irradiated by plutonium

Healing Petal [Pink Petals · CMD 1105]
Harvested from Mutated Healing Plants. Ingredient for radiation cures.
&" Primary cure ingredient

Machines & Tools

Nuclear Smelter [Blast Furnace · CMD 1201]
Industrial refining machine. Converts Raw Fragments !' Refined Ingots.
&" Right-click to open

Nuclear Forge [Smithing Table · CMD 1401]
Upgrade machine. Upgrades plutonium & hazmat gear from MK-1 to MK-4.
&" Requires Radioactive Core fuel

Radiation Drill [Diamond Pickaxe · CMD 1108]
Only tool that safely mines Plutonium Ore. Diamond+ speed.
&" No radiation burst on use

Consumables & Cures

Radiation Antidote [Honey Bottle · CMD 1301]
Clears all radiation, infection, and debuffs. Does NOT grant immunity.
&" Right-click to consume

Radiation Serum [Glass Bottle · CMD 1302]
Instantly clears radiation AND grants 10 min immunity. Rare & expensive.
&" Right-click to consume

Titan / Boss Items

Titan Core [Nether Star · CMD 1107]
Crystallized essence of the Plutonium Titan. 4 required for summoning ritual.
&" EXTREME radiation hazard

Titan Fragment [Amethyst Shard · CMD 1108]
Drop from the Titan. Used in high-tier crafting.
&" Extreme radiation

Reactor Heart [Nether Star · CMD 1109]
Rare Titan drop. Required for MK-3/4 Forge upgrades.
&" Critical radiation hazard

Mutated Crystal [Amethyst Shard · CMD 1111]
Rare Titan drop. Grown inside the Titan's body.
&" Handle with care

Ancient Reactor Blueprint [Paper · CMD 1110]
Very rare Titan drop. Unlocks Phase 11 advanced crafting.
&" Future content

&" Crafting Recipes — Machines & Tools

Nuclear Smelter

&" Crafted at a normal Crafting Table.

Nuclear Smelter

Diamond	Obsidian	Diamond
Obsidian	Furnace	Obsidian
Diamond	Obsidian	Diamond

! Produces 1× Nuclear Smelter block
! D = Diamond · O = Obsidian · F = Furnace

Nuclear Forge

&" Requires Echo Shard (= Refined Plutonium Ingot) in your inventory.

Nuclear Forge

Obsidian	Ancient Debris	Obsidian
Echo Shard	Smithing Table	Echo Shard
Obsidian	Ancient Debris	Obsidian

! Produces 1× Nuclear Forge block
! Echo Shard = Refined Plutonium Ingot (proxy material)

Radiation Drill

&" The ONLY tool that can mine Plutonium Ore safely.

Radiation Drill

R-Core	Diamond	R-Core
Diamond	Diamond	Diamond

!' Produces 1× Radiation Drill
!' R-Core = Radioactive Core · Uses Diamond Pickaxe as base

Plutonium Arrow

Plutonium Arrow (×4)

	Ref. Ingot	
	Arrow	
	Feather	

!' Ref. Ingot = Refined Plutonium Ingot (Echo Shard proxy)
!' Applies 25 radiation + Poison + Glowing on hit

&" Crafting Recipes — Weapons & Armor

All recipes below use Echo Shard as a proxy for Refined Plutonium Ingot.
P = Echo Shard (Refined Plutonium Ingot) S = Stick W = Yellow Wool (Industrial Fabric)

Plutonium Sword

	P	
	P	
	S	

! 10 DMG · +10 Rad/hit · 2,500 durability

Plutonium Pickaxe

P	P	P
	S	
	S	

! 10% chance to find extra Debris · 2,800 dur.

Plutonium Axe

P	P	
P	S	
	S	

! 11 DMG · 15% shockwave (3-block rad) · 2,600 dur.

Plutonium Shovel

	P	
	S	
	S	

! 15% soil conversion · 2,500 dur.

Plutonium Hoe

P	P	
	S	
	S	

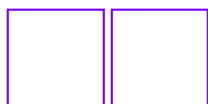
! Fast farming · 2,000 dur.

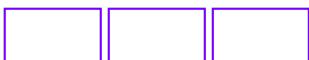
Plutonium Armor

P = Echo Shard (Refined Plutonium Ingot) · All pieces enchanted: Protection III, Unbreaking III

Plutonium Helmet

P	P	P

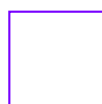




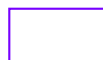
! 4 Armor · 2 Toughness · 450 dur. · -15% Radiation

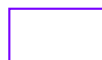
Plutonium Chestplate

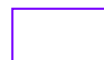












! 8 Armor · 2 Toughness · 640 dur. · -15% Radiation

Plutonium Leggings

! 6 Armor · 2 Toughness · 590 dur. · -15% Radiation

Plutonium Boots

! 4 Armor · 2 Toughness · 480 dur. · -15% Radiation · Feather Fall II

&" Crafting Recipes — Hazmat Suit & Cures

Hazmat Suit

W = Yellow Wool (Industrial Fabric proxy) · Based on Leather armor
Full set bonus: 80% radiation reduction (best non-plutonium protection)

Hazmat Helmet

W	W	W
W		W

! 2 Armor · 0.5 Tough · 363 dur. · -20% Rad

Hazmat Chestplate

W		W
W	W	W
W	W	W

! 5 Armor · 0.5 Tough · 529 dur. · -30% Rad

Hazmat Leggings

W	W	W
W		W
W		W

! 4 Armor · 0.5 Tough · 496 dur. · -20% Rad

Hazmat Boots

W		W
W		W

! 2 Armor · 0.5 Tough · 430 dur. · -10% Rad

Radiation Cures

Radiation Antidote (Shapeless Recipe)

Healing Petal	Healing Petal	Honey Bottle
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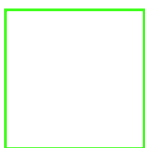
!Radiation Antidote ×1

Effect: Clears ALL radiation, infection & debuffs. No immunity granted.

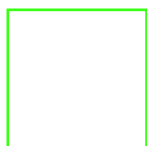
Radiation Serum (Shaped Recipe)

Radiation Serum

Healing Petal	R-Core	Healing Petal
Gold Nugget	Gold Apple	Gold Nugget



Glass Bottle



!' R-Core = Radioactive Core
!' Effect: Clears ALL radiation + grants 10 MINUTES of immunity

&" Radiation System

Radiation Stages

Stage 0 — Healthy	0–99 points	No effects. Normal gameplay.
Stage 1 — Minor Exposure	100–249 points	Weakness I + Nausea
Stage 2 — Moderate Exposure	250–499 points	Weakness I + Slowness I + Nausea · 5% spread chance to nearby players
Stage 3 — Severe Exposure	500–749 points	Weakness II + Slowness II + Hunger I · 0.5 HP damage/cycle · 15% spread chance
Stage 4 — Critical Poisoning	750–1000 points	Weakness III + Slowness III + Hunger II · 1.0 HP damage/cycle · 25% spread chance

Radiation Sources

Plutonium Ore (nearby)	+5 per check (5s interval)
Mining without Radiation Drill	+25 burst (5s cooldown)
Irradiated Zombie hit	+10 per hit
Alpha Zombie (Lv.4) hit	+50 per hit
Nuclear Smelter (nearby)	+8 per 5 seconds (3-block radius)
Radioactive Debris	+0.5/s in 5-block radius
Radiation Cloud (zombie death)	+5 per second for 10 seconds
Titan Boss Aura	+3 per second in 10-block radius
Titan Boss Slam	+15 per hit
Titan Radiation Wave	+40 (20-block radius)
Titan Reactor Overload	+70 (18-block radius)
Titan Energy Beam (targeted)	+60 (direct hit)
Titan Nuclear Catastrophe	+80 (30-block radius)
Healing Petal (direct consume)	-25 points
Radiation Antidote	Clears ALL radiation
Radiation Serum	Clears ALL + 10 min immunity

Armor Protection

	Each armor piece reduces incoming radiation by a flat percentage.
Any armor piece	-15% per piece (max 60% for full set)
Hazmat Suit — per piece	-10% to -30% per piece
Hazmat Suit — full set	-80% total radiation reduction
Plutonium Armor — full set	Full environmental radiation IMMUNITY
Plutonium Boots special	No radiation from ground contact

Contagion (Radiation Spread)

Proximity spread (Stage 2)	5% chance / 5 seconds within 3 blocks · +25 rad
Proximity spread (Stage 3)	15% chance / 5 seconds · +25 rad
Proximity spread (Stage 4)	25% chance / 5 seconds · +25 rad
Physical contact (melee hit)	35% chance · +50 rad to victim
Vehicle sharing (boat/mincart)	20% chance / 15 seconds · +25 rad

Natural Decay

Decay interval	Every 60 seconds
Grace period required	No new radiation for 10 minutes
Amount removed	-2 radiation per decay tick

&" Machines — Nuclear Smelter

How It Works

The Nuclear Smelter converts Raw Plutonium Fragments into Refined Plutonium Ingots. Right-click the placed Smelter block to open its GUI interface.

GUI Slots

Input Slot	Place Raw Plutonium Fragments here
Fuel Slot	Insert fuel items (Coal, Charcoal, Blaze Rod, Lava Bucket)
Output Slot	Collect Refined Plutonium Ingots

Fuel Values

Coal	100 fuel units (~20 seconds of processing)
Charcoal	80 fuel units
Coal Block	900 fuel units (~3 minutes)
Blaze Rod	120 fuel units
Lava Bucket	1,000 fuel units (~3.3 minutes) !• BEST

Temperature System

Min processing temp	500°C — must reach this before processing begins
Max safe temp	1,500°C — overheats above this
Heating rate	+5°C per tick cycle while fueled
Active rate	+2.5°C per tick while processing
Cooling rate	-1°C per tick while idle
Overheat cooldown	10 seconds overheated !' then cools down
Restart threshold	Machine restarts at 80% of minimum temp (avoids cold start)

Processing Recipe

Input	1× Raw Plutonium Fragment
Output	1× Refined Plutonium Ingot
Time	15 seconds (300 ticks at 20 TPS)
Fuel cost	1 fuel unit per tick cycle

Radiation Hazard

Passive radius	The Nuclear Smelter emits passive radiation while active.
Radiation per check	3 blocks around the machine
Check interval	+8 points
	Every 5 seconds (100 ticks)

&" Keep the Smelter in a dedicated room away from your base!

&" Machines — Nuclear Forge

How It Works

The Nuclear Forge upgrades Plutonium tools, weapons, and Hazmat armor to MK-I through MK-IV. Insert Radioactive Cores as fuel, then right-click to open the upgrade GUI.

		Upgrade Tiers	Success	Bonus	Fail Risk
MK-1	2 Ingots + 1 Core	500	100%	+5% damage, speed, armor, durability	
MK-2	4 Ingots + 2 Cores	1,200	90%	+10% all stats	0%

+20% all stats

MK-4

16 Ingots + 8 Cores + 2 Hearts

5,000

50%

+35% all stats

0%

5% !' MK-3

MK-IV Special — Radiation Aura

Aura radius

Effect on hostile mobs

Effect on enemy players (PvP)

Aura tick rate

Forge Overload burst

3 blocks around player

+25 radiation per aura tick

+15 radiation per aura tick

Every 2 seconds (40 ticks)

+150 radiation to nearby players in 8-block radius

&" World: Ore, Mining & Farming

Plutonium Ore

Location	Deep underground — Y: -64 to -58 (deepslate layer)
Spawn chance	60% of newly generated chunks get a vein
Vein size	1–2 blocks per vein
Host blocks	Deepslate, Stone, Cobbled Deepslate, Tuff

Mining Rules

Without Radiation Drill	+25 radiation burst (5s cooldown) — ore does NOT drop
With Radiation Drill	Safe extraction — ore drops normally, no burst
Base drop	1 Raw Plutonium Fragment per ore
Fortune I	1–2 fragments
Fortune II	1–3 fragments
Fortune III	1–4 fragments
XP per mine	3–7 XP orbs

Ore Radiation Zones

Within 1 block of ore	+5 radiation / 5 seconds
Within 2 blocks	+2 radiation / 5 seconds
Within 3 blocks	+1 radiation / 5 seconds
Hazmat suit nearby	80% of radiation blocked (×0.20 multiplier)
Plutonium armor nearby	100% blocked — complete immunity

Plutonium Block

Recipe	9× Refined Plutonium Ingots in a 3×3 grid (compact storage)
Radiation tier 1	+10 per check within 1 block
Radiation tier 2	+5 per check within 2 blocks
Radiation tier 3	+2 per check within 3 blocks

Radioactive Debris

Location	Y: -64 to -32 underground (rare)
Rarity	0.1% chance per chunk
Drop on mine	15% chance: Radioactive Core
Passive radiation	+0.5 rad/s within 5 blocks

&" Radioactive Farming

How to Farm

1. Obtain Mutated Seeds (dropped by Irradiated Zombies)
2. Find or convert farmland into Radioactive Farmland (use Plutonium Shovel — 15% chance)
3. Plant Mutated Seeds on Radioactive Farmland only
4. Let the plant grow through 5 stages (0 = seed ! 4 = mature blossom)
5. Harvest to get Healing Petals (1–3) and bonus Mutated Seeds (0–2)

Farming Stats

Growth bonus (Radioactive Petals)	+50% extra growth chance per tick
Harvest	1–3 Healing Petals

Seeds per harvest

XP per harvest
Toxic Bloom chance

0–2 Mutated Seeds (replanting supply)
2–5 XP
1% — fully-grown crop mutates instead of dropping normally

Toxic Bloom Hazard

Radiation range
Radiation amount
Tip

4-block radius around the bloom
+8 per 2 seconds (40 ticks)
Break it immediately with any tool to stop it radiating

Radioactive Farmland Radiation

Passive exposure
Hazmat boots special

+1 radiation per 2 seconds to nearby players
Completely negates ground-contact radiation

&" Irradiated Zombies

Spawning

60% of naturally spawning zombies become Irradiated Zombies. Baby zombies are excluded.

Zombie Levels & Stats	HP	DMG	Speed	KB Res	Radiation	XP
Level 1 (80%)	35 HP	5 DMG	×1.15 speed	0.15 KB resist	+10 rad/hit	10 XP
Level 2 (15%)	45 HP	7 DMG	×1.20 speed	0.20 KB resist	+20 rad/hit	20 XP
Level 3 (4%)	60 HP	9 DMG	×1.25 speed	0.30 KB resist	+35 rad/hit	40 XP
Level 4 Alpha (1%)	80 HP	12 DMG	×1.30 speed	0.50 KB resist	+50 rad/hit	100 XP

&" Level 4 Alpha: has green Bossbar, glows, visible from afar. Extremely dangerous.

	Lv.1	Lv.2	Lv.3	Lv.4 Alpha
Rotten Flesh	0–3 always	0–3 always	0–3 always	0–3 always
Radioactive Core	15%	25%	40%	100% (always!)
Mutated Seed	5%	10%	20%	50%
Irradiated Heart	1%	3%	7%	25%

Radiation Cloud (On Death)

Spawn chance	20% per zombie death
Cloud radius	3 blocks
Duration	10 seconds
Radiation	+5 per second to players inside

Radiation Surge Night Event

Chance per night	5%
Effect	ALL radiation damage doubled, ALL loot drops doubled
Duration	One full in-game night
Warning	Bossbar shown to all players: &" RADIATION SURGE ACTIVE &"

Combat System vs. Zombies

Plutonium Sword vs zombie	+10 radiation per hit on zombie
Axe shockwave (15% chance)	3-block radius · +15 radiation to all in area
Plutonium Arrow hit	+25 radiation · Poison effect · Glowing
MK-IV Aura	Passive +25 radiation per 2s to nearby mobs

&" Plutonium Titan Boss

Summoning Ritual

Build the Summoning Altar with Crying Obsidian at the center and 4 Obsidian blocks at each corner (2 blocks away diagonally). Stand at the center and:

1. Hold 4× Titan Cores in your inventory
2. Right-click the center Crying Obsidian with an Irradiated Heart
3. All 4 Titan Cores are consumed immediately
4. A 15-second summoning animation begins
5. The Plutonium Titan spawns at the altar center

&" 60-minute cooldown between summons. Only one Titan can be alive at a time.

Health Scaling (by players nearby)

1–3 players nearby	5,000 HP
4–6 players nearby	6,500 HP
7–10 players nearby	8,500 HP
10+ players nearby	10,000 HP

Phase System

Phase 1 (100% – 75% HP)	×1.0 speed, ×1.0 damage	Titan Slam · Radiation Aura
Phase 2 (75% – 50% HP)	×1.1 speed, ×1.2 damage	Radiation Wave · Mutant Summoning · Aura
Phase 3 (50% – 25% HP)	×1.25 speed, ×1.5 damage	Reactor Overload · Energy Beam · All above
Phase 4 (25% – 0% HP)	×1.5 speed, ×2.0 damage	Nuclear Catastrophe · Final Frenzy · All above

All Titan Abilities

Ability	Phase	Details
Radiation Aura (Passive)	ALL Phases	+3 rad/sec in 10-block radius at all times
Titan Slam	Phase 1+	5-block radius · 12 base DMG · +15 radiation · knockback · 10s CD
Radiation Wave	Phase 2+	20-block expanding ring · +40 radiation · 15s CD
Mutant Summoning	Phase 2+	Summons 3 regular + 1 Alpha zombie · 25s CD
Reactor Overload	Phase 3+	5-second charge · 18-block radius · +70 radiation · 40s CD
Energy Beam	Phase 3+	Targets random player · 25 DMG + 60 radiation · 20s CD
Nuclear Catastrophe	Phase 4 only	10-second charge · 30-block radius · sets HP to 1 · +80 rad · 90s CD
Final Frenzy	Phase 4 only	One-time trigger · +50% attack speed permanently for rest of fight

Arena Hazard

Hazard radius	15 blocks from Titan spawn point
Radiation inside arena	+2 radiation per check
Visual	Particle effects mark the arena boundary

Titan Rewards

XP on kill	5,000 XP (split by contribution)
Titan Fragment (guaranteed)	1–8+ depending on % contribution
Refined Plutonium Ingots	Scales with contribution ("e5%)
Reactor Heart (rare)	10% chance per qualifying player
Ancient Reactor Blueprint	5% chance per qualifying player
Mutated Crystal	8% chance per qualifying player
Loot Chest	Spawns at death location with bonus Cores, Fragments & Ingots

&" Combat System & Progression

Weapon Radiation Effects

Plutonium Sword (base)	+10 radiation per hit
Sword critical hit (falling)	+5 bonus radiation
Sword vs high-stage player	+5 bonus when victim is Stage 3+
Plutonium Axe (base)	+11 DMG
Axe shockwave (15% chance)	3-block radius radiation burst · +15 rad
Plutonium Arrow (hit)	+25 radiation · Poison (4s) · Glowing (5s)
Arrow critical (full charge)	+20 bonus radiation
Arrow headshot bonus	+10 radiation

Combo System

Max combo stack	8 hits
Combo reset time	6 seconds since last hit
Bonus per stack	+5 extra radiation per hit above 1
Maximum bonus	+35 radiation from combo (capped)

Radiation Surge (PvP)

Trigger condition	Both fighters must be Stage 2+ radiation
Trigger chance	5% per hit
Effect on both fighters	+30 radiation each
Effect on bystanders	+15 radiation (8-block radius)
Cooldown	15 seconds per pair

Weapon Mastery System

Novice (0 XP)	Starting level
Experienced (100 XP)	Sword: +1 XP/hit, +10/kill
Veteran (500 XP)	Axe: +1 XP/hit, +10/kill
Elite (1,500 XP)	Bow: +2 XP/hit, +15/kill
Master (4,000 XP)	+10% radiation bonus on all weapon hits

Equipment Repair

Plutonium tools & armor	Repair with Refined Plutonium Ingot (+300 durability per ingot)
Hazmat Suit	Repair with Industrial Fabric (+200 durability per piece)
Method	Use vanilla Anvil with the repair material

&" Admin Commands

<code>/nc radiation set <player> <amount></code>	Set a player's radiation level directly
<code>/nc radiation add <player> <amount></code>	Add radiation to a player
<code>/nc radiation clear <player></code>	Clear all radiation from a player
<code>/nc radiation status <player></code>	Check a player's radiation stage and level
<code>/nc ore give fragment [amount]</code>	Give Raw Plutonium Fragments
<code>/nc ore give drill</code>	Give a Radiation Drill
<code>/nc equipment give <type></code>	Give plutonium/hazmat gear (sword, axe, helmet, etc.)
<code>/nc farming give <seed petal antidote serum></code>	Give farming/cure items
<code>/nc smelter give</code>	Give a Nuclear Smelter block
<code>/nc forge give</code>	Give a Nuclear Forge block
<code>/nc titan spawn</code>	Force-spawn the Plutonium Titan at your location
<code>/nc titan kill</code>	Force-kill the active Titan
<code>/nc titan phase <1-4></code>	Force the Titan to a specific phase

/nc reload

Reload all plugin configs

1.21+ · Phases 1–10 Complete

All values are configurable via the YAML config files in the plugin folder.

